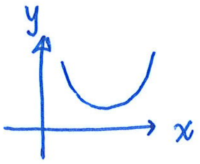
1 The equation of a curve is $y = kx^2 + kx + p$, where p and k are constants.

(a) Show that $p > \frac{k}{4}$ for which the curve lies completely above the x-axis [3]



$k > 0$ &

no real roots

$\Rightarrow$ Discriminant < 0

$b^2 - 4ac < 0$

$(k)^2 - 4(k)(p) < 0$ M1

$k^2 - 4kp < 0$

$k(k - 4p) < 0$

Since $k > 0$

$\therefore\ k - 4p < 0$ M1

$k < 4p$

$4p > k$

$p > \frac{k}{4}$ (shown) A1

(b) In the case where $k = 2$ and $p = 4$, find the values of $\boldsymbol{m}$ for which the line $y = mx - 4$ is a tangent to the curve. [4]

$y = 2x^2 + 2x + 4$ ——— ①

$y = mx - 4$ ——— ②

① = ②,

$2x^2 + 2x + 4 = mx - 4$

$2x^2 + (2 - m)x + 8 = 0$ M1

For line is a tangent to the curve,

discriminant, $D = 0$

$\Rightarrow b^2 - 4ac = 0$ M1

$(2-m)^2 - 4(2)(8) = 0$

$4 - 4m + m^2 - 64 = 0$

$m^2 - 4m - 60 = 0$

$(m - 10)(m + 6) = 0$ → Minus 1 Mark if this step not shown

$m = 10$ or $m = -6$ A2